UNIVERSITY OF IBADAN

FACULTY OF TECHNOLOGY

DEPARTMENT OF AGRICULTURAL AND ENVIRONMENTAL ENGINEERING

B.Sc. Degree Examination

First Semester, 2016/2017 Session

July 2017

TAE 412: Engineering Properties of Plant and Animal Materials (3 Units)

Time Allowed: 3 hours

Instruction:

Attempt all the questions

AOR

QUESTION ONE (30 Marks)

(i) Explain briefly the appropriate terminologies and their associated time parameters, for determining the stack height and safe period of handling, for agricultural products especially when transporting over very rough roads. **(8 marks)**

(ii) Using appropriate rheological models and the conditions of loading/deformation in (i), proof, from the first principles,

a. the definitions of the appropriate time parameters associated with the terminologies **(20 marks)**

b. that biomaterials are time dependent **(2 marks)**.

QUESTION TWO (30 marks)

i. Discuss, for each of the followings, two methods for
   a. determination and
   b. computation of moisture content in biomaterials. **(6 marks)**

ii. A feed mill obtained maize from two sources (50 tonnes each), one has 12% moisture content (mc) dry basis and the other 12% mc wet basis. Which one has more dry matter? Justify your answer with calculations. **(6 marks)**

iii. Give a brief explanation of moisture sorption and explain the appropriate terms. Give a sketch of the moisture isotherm for any three temperatures and what happens during repeated sorptions at any of the temperatures. **(8 marks)**

iv. Define the terms in the equations below. Proof that as the material approaches the liquid state μ approaches 0.5.

$\frac{1}{E}=\frac{1}{3G}+\frac{1}{9K}$, $\quad E=3K(1-2\mu)$, $\quad E=2G(1+\mu)$ **(4 marks)**

v. Describe (with appropriate diagram) a method for determining the porosity of granular materials that has the capability of absorbing water. **(6 marks)**

QUESTION THREE (20 marks)

(i) Explain briefly the four classes of non-Newtonian fluids and give two examples for each. Use equations and graphics where possible. **(8 marks)**

(ii) Briefly describe any two types of viscometers (use Equation(s) and sketch(es) to support your answer. **(4 marks)**

(iii) The data below was obtained for a non-Newtonian fluid in a capillary viscometer 20cm long and 3cm wide using. Determine the viscous coefficient and the fluid index. **(8 marks)**

$$\frac{\Delta P.D}{4L}=\tau=K\left(\frac{8V}{D}\right)^n$$

Velocity (m/s)	0.0055	0.0071	0.0098	0.0128	0.0162	0.0216	0.0260	0.0300
ΔP (N/m^2)	157.02	211.82	277.33	362.67	461.33	613.33	738.67	853.33

QUESTION FOUR (20 Marks)

i. Give one example and one significance in engineering, for each of the following: Physical, Mechanical, Electrical and Thermal properties of biomaterials. **(8 marks)**

ii. Briefly explain with graphics where possible
   a. the basic characteristics of biomaterials
   b. the ways in which water occur in biomaterial **(6 marks)**

iii. A sample of feed weighing 250g was shaken on a ROTAP machine containing seven sieves with a pan. The table below gives the weight of the particles retained on the sieves. Determine the two indices often used to express the particle size distribution. What are the implications of the values obtained?

Sieve/screen No	3/8	4	8	14	28	48	100	Pan
Weight of feed	2.50	6.25	17.50	60.00	88.75	56.25	18.75	0

(6 marks)